

作业 2

Problem 1

$$(1) \frac{2z - i}{iz + z}, \quad z = x + iy$$

$$\Rightarrow \frac{2(x+iy) - i}{(-y+ix) + z} = \frac{2(x+iy) - i}{(2-y) + ix}$$

$$= \frac{(2(x+iy) - i)(12-y) - ix}{(2-y)^2 + x^2}$$

$$\Rightarrow (2x + i2y - i)(12-y) - ix$$

$$= (2x + i(2y-1))(12-y) - ix$$

$$= 2x(2-y) + i(2y-1)(2-y)$$

$$+ i(-x)(2x) + x(2y-1)$$

$$\Rightarrow \frac{2x(2-y) + x(2y-1) + i((2y-1)(2-y) + (-x)(2x))}{(2-y)^2 + x^2}$$

$$u(x, y) = \frac{4x - 2xy + 2xy - x}{(2-y)^2 + x^2} = \frac{3x}{(2-y)^2 + x^2}$$

$$v(x, y) = \frac{-2y^2 + y + 4y - 2 - 2x^2}{(2-y)^2 + x^2} \quad \square = (2-y)^2 + x^2$$

$$\frac{\partial u}{\partial x} = \frac{1}{\square^2} (3((2-y)^2 + x^2) - 3x(2x))$$

$$= \frac{1}{\square^2} (3(2-y)^2 - 3x^2)$$

$$= \frac{3}{\square^2} ((2-y)^2 - x^2)$$

$$\frac{\partial u}{\partial y} = \frac{1}{\square^2} (0 + 3x(2(2-y)))$$

$$= \frac{6}{17^2} x (2-y)$$

$$\frac{\partial v}{\partial x} = \frac{1}{17^2} (-4x(2-y)^2 + x^2) - (-2y^2 + 5y - 2 - 2x^2) 2x$$

$$= \frac{1}{17^2} (-4x(2-y)^2 - 4x^3 + 4x^3 - 2x(-2y^2 + 5y - 2))$$

$$= \frac{2}{17^2} (-2x(y^2 - 4y + 4) + x(2y^2 - 5y + 2))$$

$$= \frac{2}{17^2} (-2xy^2 + 2xy^2 + 8xy - 8x - 5xy + 2x)$$

$$= \frac{2}{17^2} (3xy - 6x)$$

$$= \frac{6}{17^2} x (y-2)$$

$$\frac{\partial v}{\partial y} = \frac{1}{17^2} (1-4y+5) (12-y)^2 + x^2 + (-2y^2+5y-2-2x^2) 2(2-y)$$

$$= \frac{1}{17^2} (-4y(12-y)^2 + 5(12-y)^2 - 4x^2y + 5x^2 + (-2y^2+5y-2)(4-2y) - 2x^2(4-2y))$$

$\begin{aligned} &\nearrow -8y + 4y^2 + 10 - 5y \\ &\nearrow -4y^2 + 10y - 4 \end{aligned}$

$$= \frac{1}{17^2} (-3x^2 + (2-y)(-4y(12-y) + 5(12-y) - 4y^2 + 10y - 4))$$

$$= \frac{1}{17^2} (3(12-y)^2 - 3x^2)$$

$$= \frac{3}{17^2} ((12-y)^2 - x^2)$$

From Cauchy-Riemann conditions

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{3}{(2-y)^2 + x^2} ((2-y)^2 - x^2)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = \frac{6}{(2-y)^2 + x^2} x(2-y)$$

so function (1) is analytic

$$(2) \quad z^2 \bar{z}, \quad z = x + iy, \quad \bar{z} = x - iy$$

$$(x+iy)^2 (x-iy) = (x^2 + i2xy - y^2)(x-iy)$$

$$= x^3 + i \cdot 2x^2y - xy^2$$

$$- i x^2y + 2xy^2 + iy^3$$

$$= (x^3 + xy^2) + i(x^2y + y^3)$$

$$u(x,y) = x^3 + xy^2, \quad v(x,y) = y^3 + x^2y$$

$$\frac{\partial u}{\partial x} = 3x^2 + y^2$$

$$\frac{\partial u}{\partial y} = 2xy$$

$$\frac{\partial v}{\partial x} = 2xy$$

$$\frac{\partial v}{\partial y} = 3y^2 + x^2$$

$$\Rightarrow \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x}$$

so function (2) is not analytic

$$(3) \sqrt{z}, \quad z = r \cdot e^{i\theta}$$

$$\sqrt{z} = \sqrt{r} \cdot e^{i\frac{\theta}{2}}$$

$$= \sqrt{r} \left(\cos \frac{\theta}{2} + i \cdot \sin \frac{\theta}{2} \right)$$

$$= \sqrt{r} \cdot \cos \frac{\theta}{2} + i \cdot \sqrt{r} \cdot \sin \frac{\theta}{2}$$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$u(r, \theta) = \sqrt{r} \cdot \cos \frac{\theta}{2}$$

$$v(r, \theta) = \sqrt{r} \cdot \sin \frac{\theta}{2}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x}$$

$$= \frac{\partial u}{\partial r} \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\partial u}{\partial \theta} \left(-\frac{y}{x^2 + y^2} \right)$$

$$= \cos \frac{\theta}{2} \cdot \frac{1}{2\sqrt{r}} \cdot \frac{x}{\sqrt{x^2 + y^2}} + \frac{\sqrt{r}}{2} (-\sin \frac{\theta}{2}) \left(-\frac{y}{r^2} \right)$$

$$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial v}{\partial \theta} \frac{\partial \theta}{\partial y}$$

$$= \sin \frac{\theta}{2} \frac{\sqrt{r}}{2r} \cdot \frac{y}{r} + \frac{\sqrt{r}}{2} (\cos \frac{\theta}{2}) \left(\frac{x}{r^2} \right)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y}$$

$$= \frac{\cos \frac{\theta}{2}}{2} \cdot \frac{\sqrt{r}}{r} \cdot \frac{y}{r} + \frac{\sin \frac{\theta}{2}}{2} (-\sqrt{r}) \left(-\frac{x}{r^2}\right)$$

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial v}{\partial \theta} \cdot \frac{\partial \theta}{\partial x}$$

$$= \frac{\sin \frac{\theta}{2}}{2} \cdot \frac{\sqrt{r}}{r} \cdot \frac{x}{r} + \frac{\cos \frac{\theta}{2}}{2} \sqrt{r} \left(-\frac{y}{r^2}\right)$$

$$\Rightarrow \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

so function (3) is analytic

$$(4) e^{iz}, \quad z = x + iy$$

$$e^{iz} = e^{-y+ix} = e^{ix} \cdot e^{-y}$$

$$e^{ix} = \cos x + i \cdot \sin x$$

$$e^{iz} = (\cos x + i \sin x) e^{-y}$$

$$u(x, y) = e^{-y} \cdot \cos x$$

$$v(x, y) = e^{-y} \cdot \sin x$$

$$\frac{\partial u}{\partial x} = e^{-y} (-\sin x)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial v}{\partial y} = e^{-y} (-\sin x)$$

$$\frac{\partial u}{\partial y} = e^{-y} (-\cos x)$$

$$\Rightarrow \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial v}{\partial x} = e^{-y} (\cos x)$$

function (4) is analytic

Problem 2

$$(1) \quad u(x, y) = 3x^2y - y^3$$

$$\frac{\partial^2 u}{\partial x^2} = 6y, \quad \frac{\partial^2 u}{\partial y^2} = -6y$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

satisfies Laplace's equation

$$v(x, y) = \int \left(-\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \right)$$

$$\frac{\partial u}{\partial y} = 3x^2 - 3y^2, \quad \frac{\partial u}{\partial x} = 6xy$$

$$v(x, y) = \int [(3y^2 - 3x^2) dx + 6xy dy]$$

$$= 3xy^2 - x^3 + C$$

$$\frac{\partial^2 v}{\partial x^2} = -6x, \quad \frac{\partial^2 v}{\partial y^2} = 6x$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

$$(2) \quad u(x, y) = e^x \cdot \cos y$$

$$\frac{\partial^2 u}{\partial x^2} = e^x \cdot \cos y, \quad \frac{\partial^2 u}{\partial y^2} = -e^x \cdot \cos y$$

$$\nabla^2 u = 0 \Rightarrow \text{satisfies Laplace's equation}$$

$$v(x, y) = \int \left(-\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \right)$$

$$= \int (e^x \sin y dx + e^x \cos y dy)$$

$$\Rightarrow v(x, y) = e^x \cdot \sin y + C$$

$\nabla^2 v = 0$ satisfy Laplace's equation

$$(3) \quad u(x, y) = \ln(x^2 + y^2)$$

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{(x^2 + y^2)^2} (2(x^2 + y^2) - 4x^2)$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{1}{(x^2 + y^2)^2} (2(x^2 + y^2) - 4y^2)$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

satisfies the Laplace's equation

$$v(x,y) = \int \left(-\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \right)$$

$$= \int \left(-\frac{2y}{x^2+y^2} dx + \frac{2x}{x^2+y^2} dy \right)$$

$$= 2 \tan^{-1} \frac{y}{x} + C$$

$$\Rightarrow \nabla^2 v = 0, \text{ satisfies Laplace equation}$$

Problem 3 $\int_1^i z dz$

a) $z = r \cdot e^{i\theta}, r = 1, d\theta$

$$dz = i \cdot (r e^{i\theta}) d\theta$$

$$\Rightarrow i \int_0^{\frac{\pi}{2}} (r e^{i\theta})^2 d\theta$$

$$= i \int_0^{\frac{\pi}{2}} e^{zi\theta} d\theta$$

$$= i \left[\frac{1}{zi} e^{zi\theta} \right]_0^{\frac{\pi}{2}}$$

$$= i \left(\frac{e^{i\pi}}{zi} - \frac{1}{zi} \right)$$

$$= \frac{e^{i\pi} - 1}{2}$$

since $e^{i\pi} = \cos\pi + i\sin\pi = -1$

$$\Rightarrow \int_1^i z dz = \frac{-1-1}{2} = -1$$

b) ① $L_1: z = 1 + it, t: 0 \rightarrow 1$

$$\int_{L_1} z dz = \int_0^1 (1+it) i dt$$

$$= i \left[t + i \frac{1}{2} t^2 \right]_0^1$$

$$= i - \frac{1}{2}$$

$$\textcircled{2} \quad z = t + i, \quad t: 1 \rightarrow 0$$

$$\int_{L_2} z dz = \int_1^0 (t + i) dt$$

$$= \left[\frac{1}{2} t^2 + it \right]_1^0$$

$$= -\frac{1}{2} - i$$

$$\int_1^i z dz = \int_{L_1} z dz + \int_{L_2} z dz$$

$$= i - \frac{1}{2} - \frac{1}{2} - i$$

$$= -1$$

Problem 4

$f(z)$ analytic, $G: |z| = 1$

$$\int_0^{2\pi} e^{i\theta} \cdot f(e^{i\theta}) d\theta = 0$$

$$\int_G z f(z) d\theta$$

$$\Rightarrow z = e^{i\theta}, dz = ie^{i\theta} d\theta$$

$$\int_G z f(z) \frac{1}{iz} dz$$

$$= \int_G \frac{f(z)}{i} dz$$

$$= -i \int_G f(z) dz$$

since $f(z)$ analytic, $\int_G f(z) dz = 0$

$$= 0$$

Problem 5

$$\oint_C \frac{\sin z}{2z - \pi} dz$$

$$a) |z| = 1, \oint_C \frac{\sin z}{2(z - \frac{\pi}{2})} dz$$

$$|z| = 1 < \frac{\pi}{2}, \text{ inside loop}$$

$$\Rightarrow \frac{\sin z}{2(z - \frac{\pi}{2})} \text{ analytic in } |z| = 1$$

$$\text{so } \oint_C \frac{\sin z}{2z - \pi} dz = 0$$

$$b) |z| = 2$$

$$\oint_C \frac{\sin z}{2z - \pi} dz = \oint_C \frac{\sin z}{2(z - \frac{\pi}{2})} dz$$

$$= \frac{1}{2} \cdot 2\pi i \sin\left(\frac{\pi}{2}\right)$$

$$= \pi i$$

Problem 6

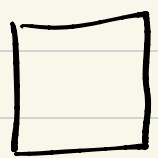
$$\oint_C \frac{\sin z}{6z - \pi} dz, |z| = 3$$

$$\oint_C \frac{\sin z}{6(z - \frac{\pi}{6})} dz, |z| = 3 > \frac{\pi}{6}$$

$$= \frac{1}{6} \cdot 2\pi i \sin\left(\frac{\pi}{6}\right)$$

$$= \frac{\sqrt{3}}{6} \pi i$$

Problem 7

$$\oint_C \frac{e^{3z}}{z - \ln^2} dz, C$$


-1-i

$$= 2\pi i \cdot e^{3\ln^2}$$

$$= 16\pi i$$

Problem 8

$$(1) \oint_{|z|=2} \frac{z^2-1}{z^2+1} dz$$

$$\frac{z^2-1}{z^2+1} = 1 + \frac{2i^2}{z^2+1}$$

$$= 1 + \frac{2i^2}{(z+i)(z-i)}$$

$$= 1 + i \left(\frac{1}{z-i} - \frac{1}{z+i} \right)$$

$$\oint_{|z|=2} \frac{z^2-1}{z^2+1} dz = \oint dz + i \int \frac{1}{z-i} dz - i \int \frac{1}{z+i} dz$$

$$= 0 + i \cdot 2\pi i - i \cdot 2\pi i$$

$$= 0$$

$$12) \oint_{|z|=2} \frac{\sin(e^z)}{z-0} dz, \quad |z|=2 > 0$$

$$= 2\pi i \cdot \sin 1$$

$$13) \oint_{|z|=2} \frac{\sin z}{z^4} dz$$

$$= 2\pi i \left(\frac{\sin z}{z^3} \right) \Big|_{z=0}$$

$$= -\frac{\pi}{3} i$$

$$14) \oint_{|z|=2} \frac{dz}{z^2(z^2+16)}$$

$$= \frac{1}{16} \left(\oint_{|z|=2} \frac{1}{z^2} dz - \oint_{|z|=2} \frac{1}{z^2+16} dz \right)$$

$$= \frac{1}{16} \left(\oint_{|z|=2} \frac{1}{z^2} dz - 0 \right)$$

$$= 0$$